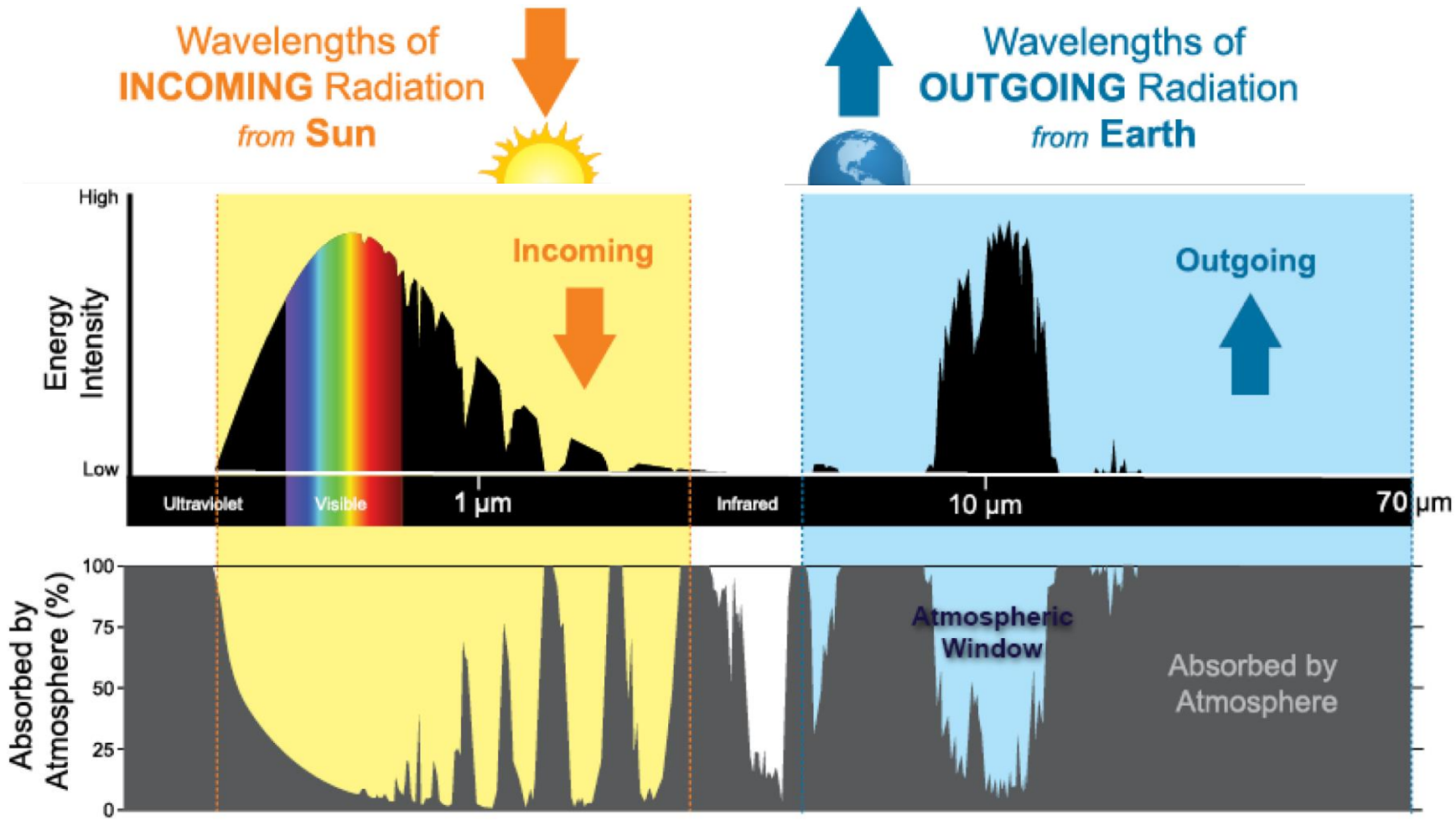


ASTR 302 Fall 2026
Earth’s Atmosphere and Adaptive Optics

Agenda – October 21, 2026

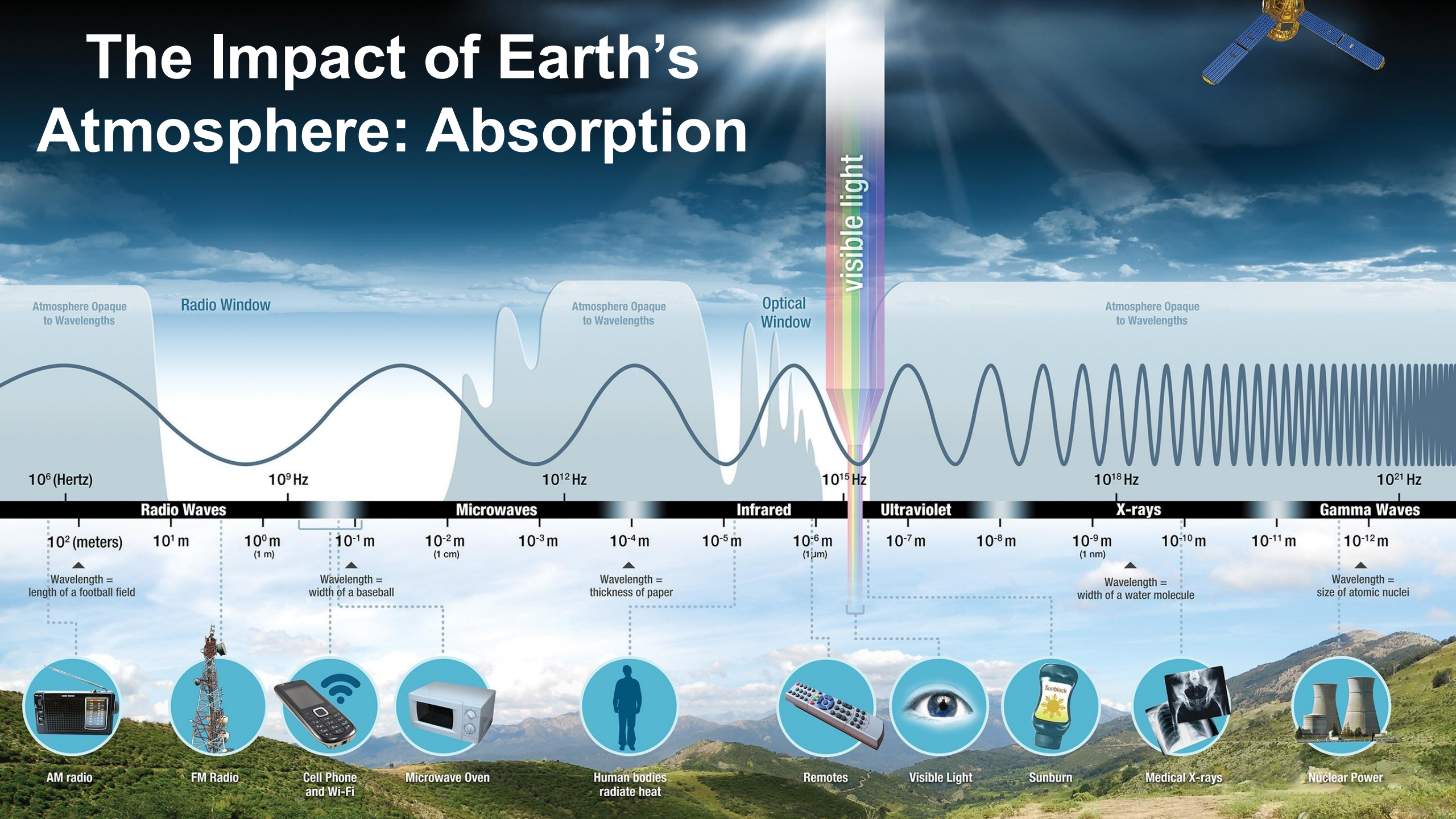
- Observing Discussion
- Dr. Miles will be in class next week (no MMT run)
- Earth's Atmosphere
- Adaptive Optics

The Impact of Earth's Atmosphere: Sky Emission

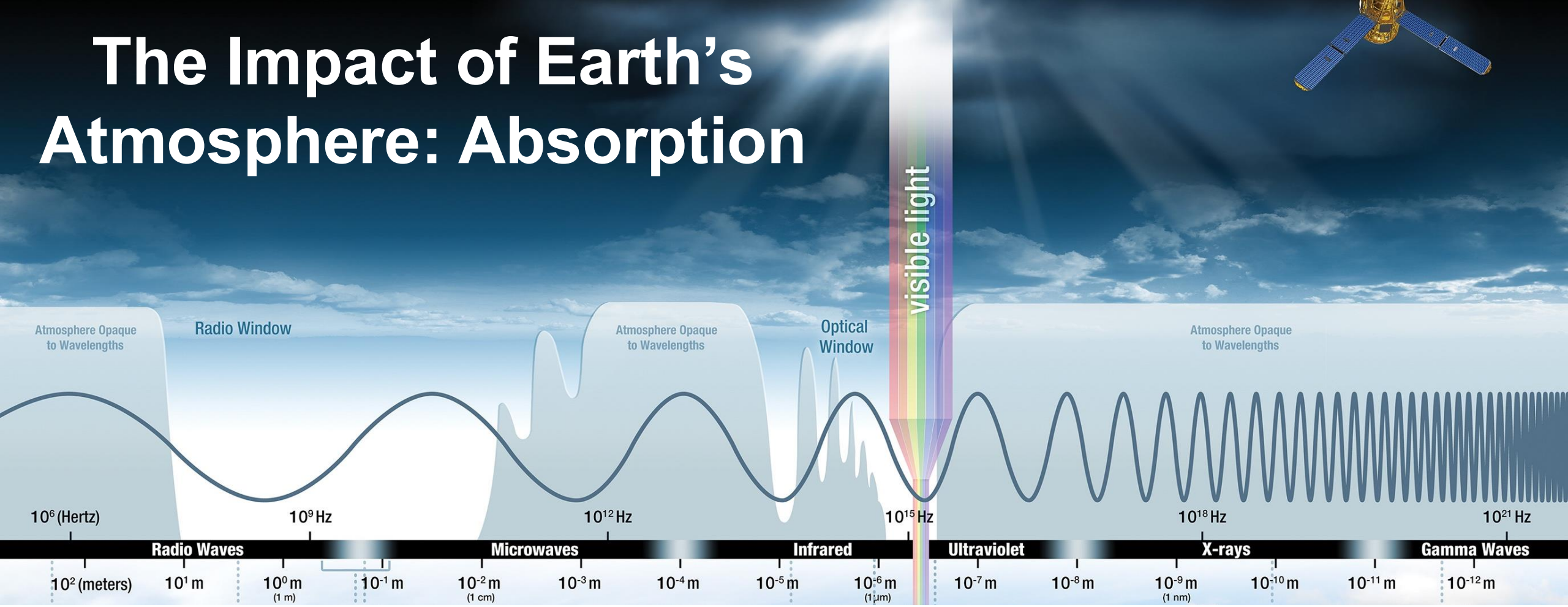


Scattering (**Optical**) and emission (**IR**) from the sky contribute to background signal in our observations.

The Impact of Earth's Atmosphere: Absorption



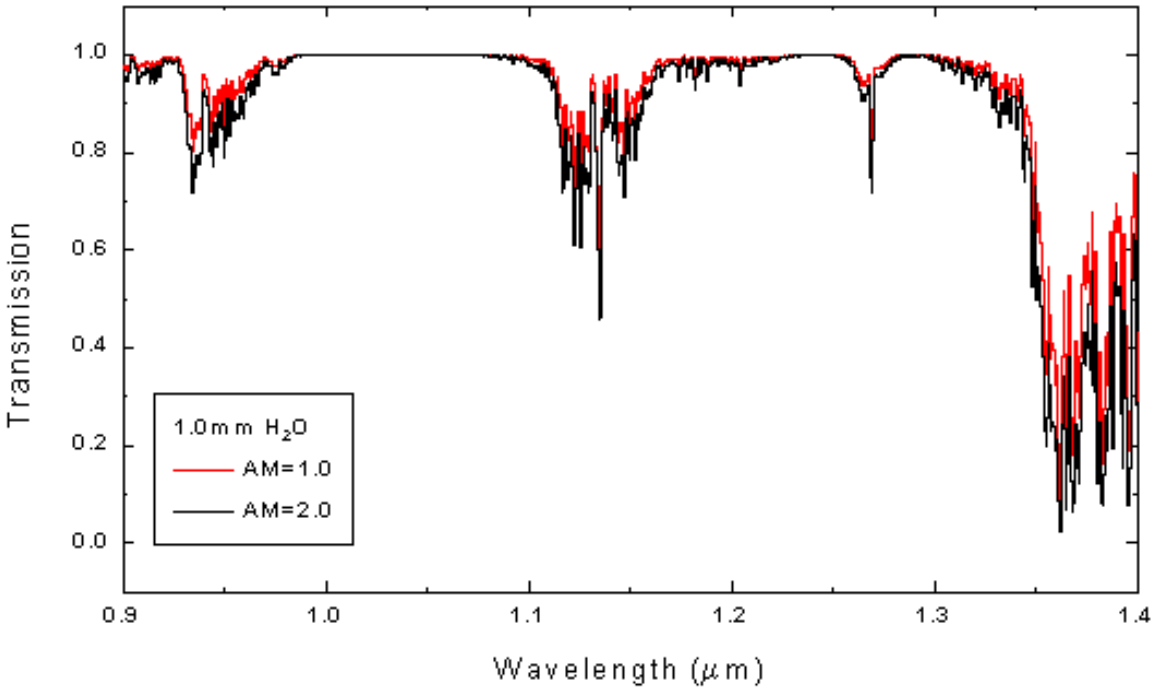
The Impact of Earth's Atmosphere: Absorption

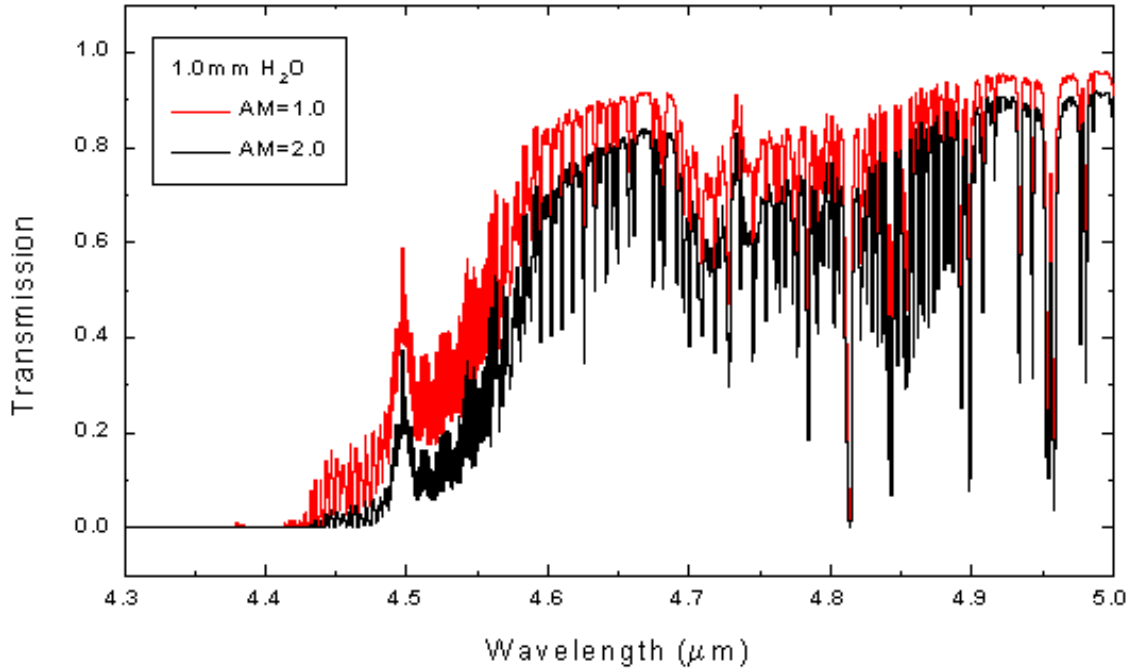


- Wavelengths significantly impacted by Earth's absorption benefit heavily from space-based observatories.

- High energy observations (**UV, X-ray, Gamma Rays**) typically require space-based or significantly elevated observatories.

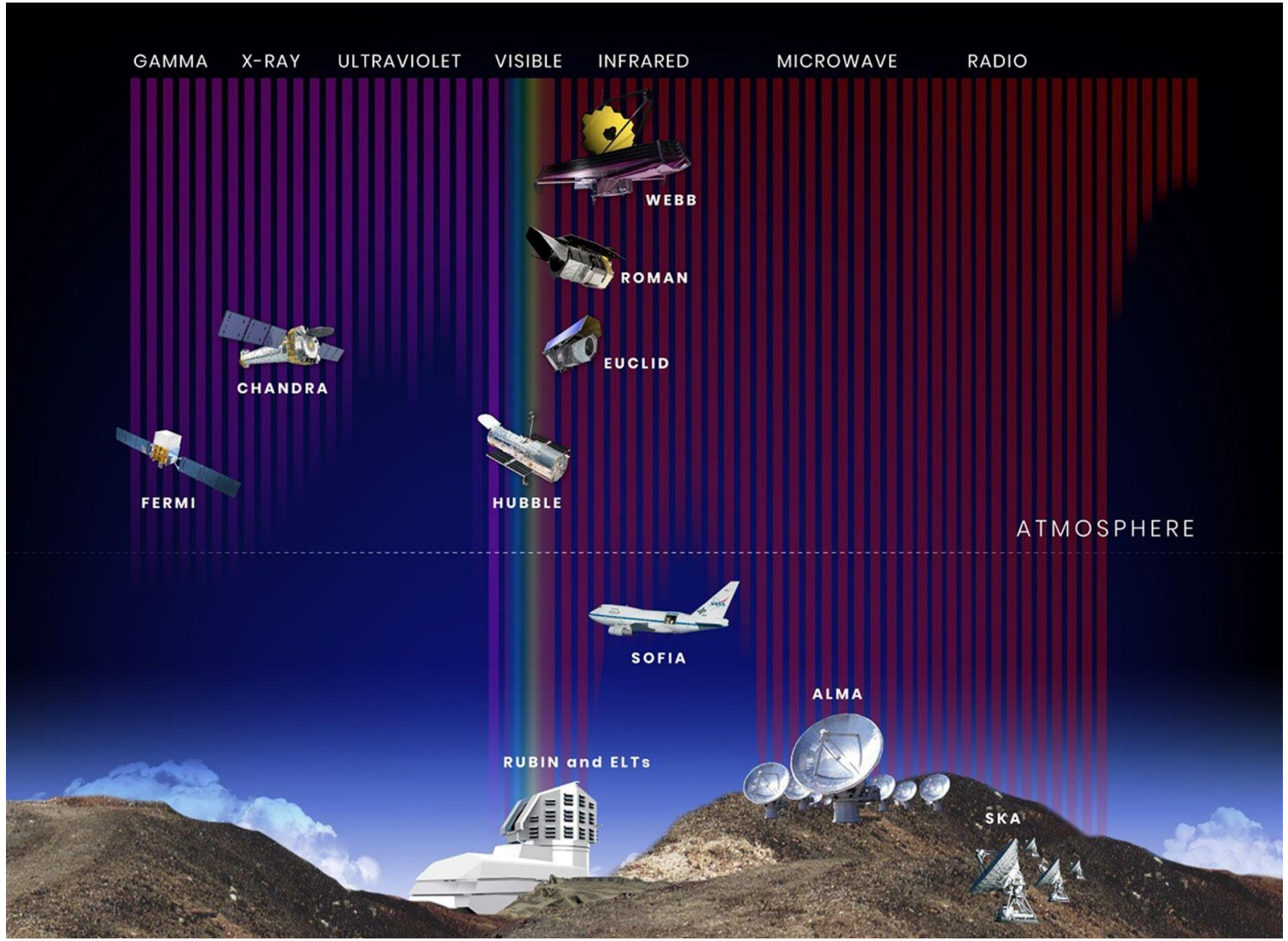
The Impact of Earth's Atmosphere: Absorption





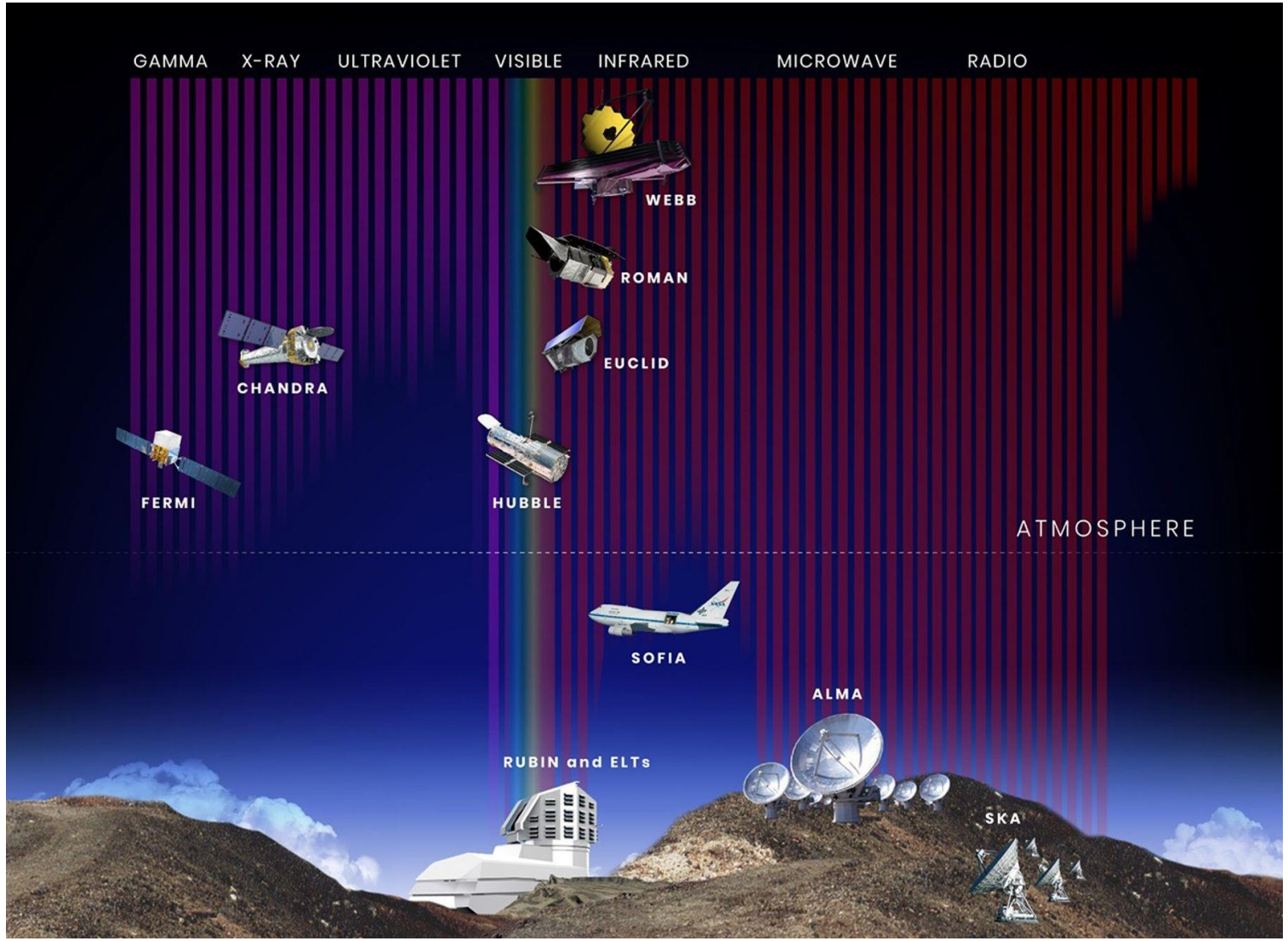
- How much atmosphere (AM) you are looking through impacts transmission.
  - Lower airmass is better for observing
  - Higher elevations are better for ground-based telescopes.

Dealing with Absorption: Going Higher



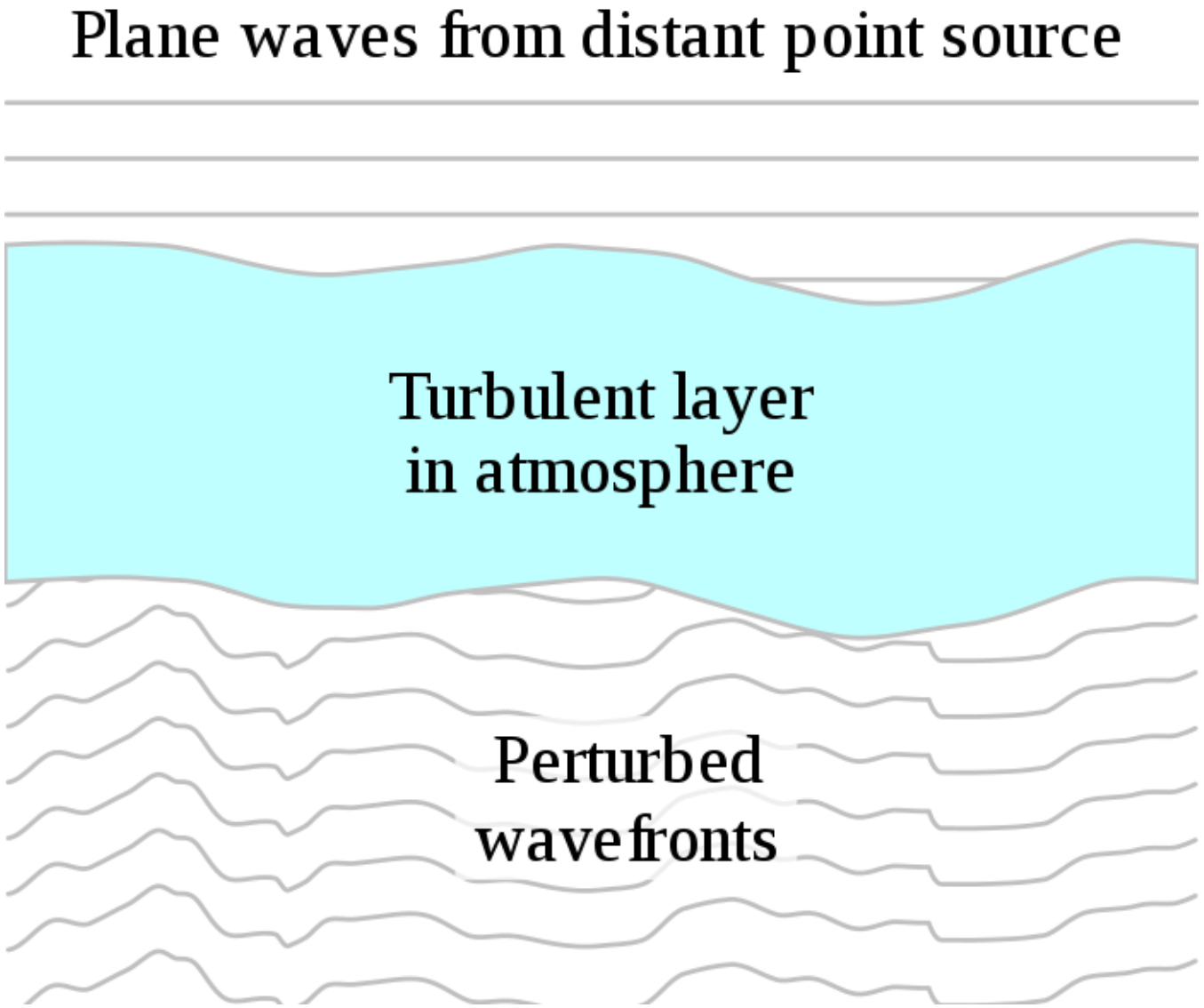
- Balloons and Planes are suborbital ways to get better access to IR and UV wavelengths.
- Hubble is in orbit but was serviceable.
- JWST is too far to be serviced by space shuttle.

Dealing with Absorption: Going Higher



- Space-based observatories are a significant public investment. They often have/had dedicated funding through Congress.
- Philanthropy has a large role in astronomy. Example: Eric and Wendy Schmidt Observatory System.

The Impact of Earth's Atmosphere: Turbulence

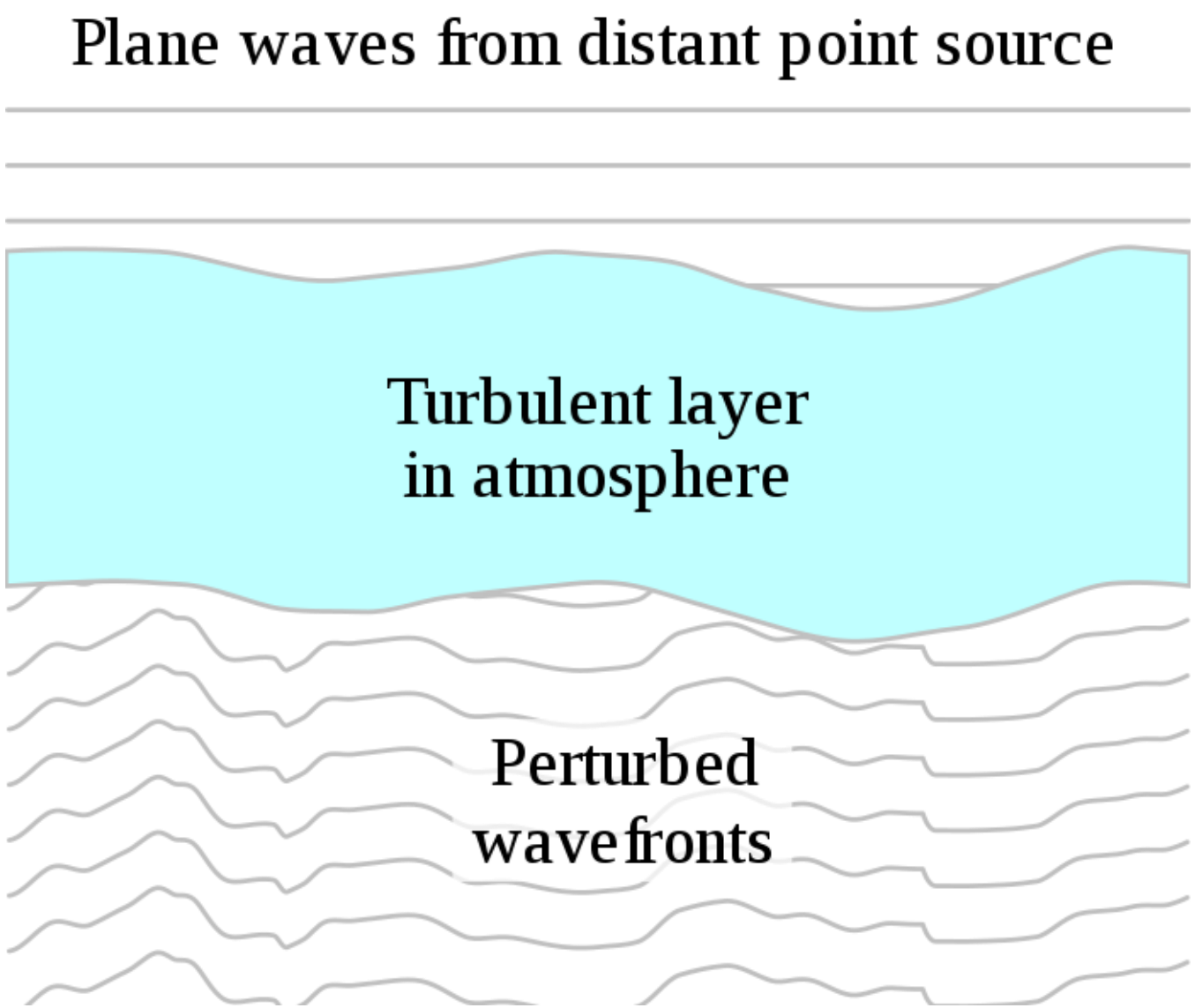


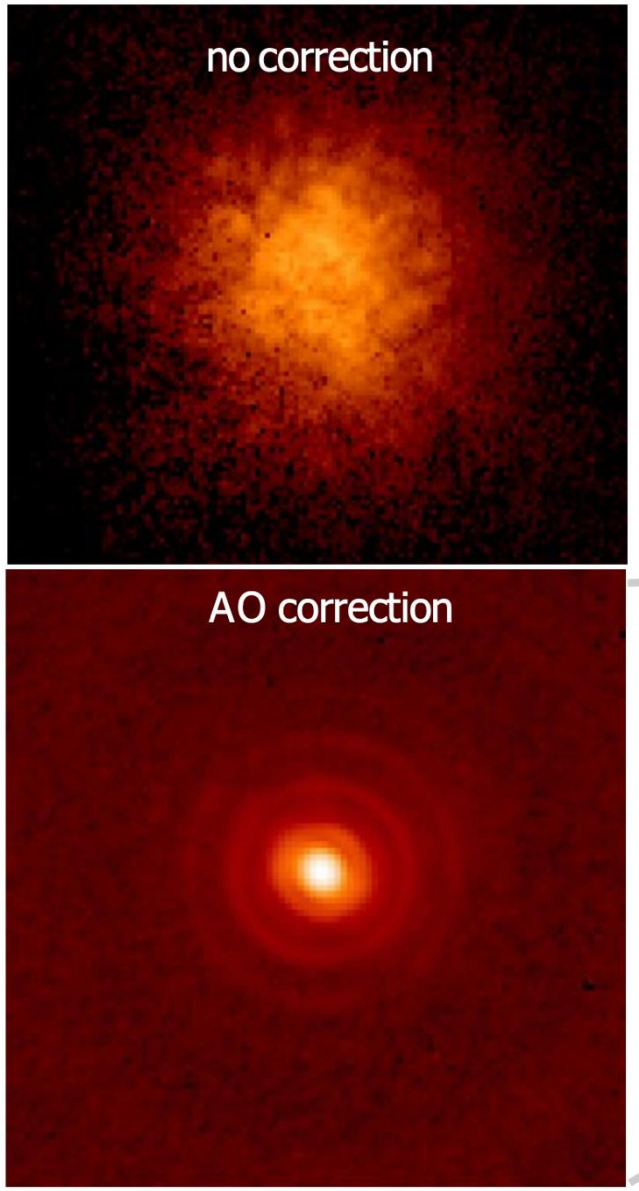
Turbulence is the primary reason telescopes do not automatically have diffraction limited spatial resolution.

Turbulence is caused by non-uniform:

- Wind
- Convection
- Temperature
- Density

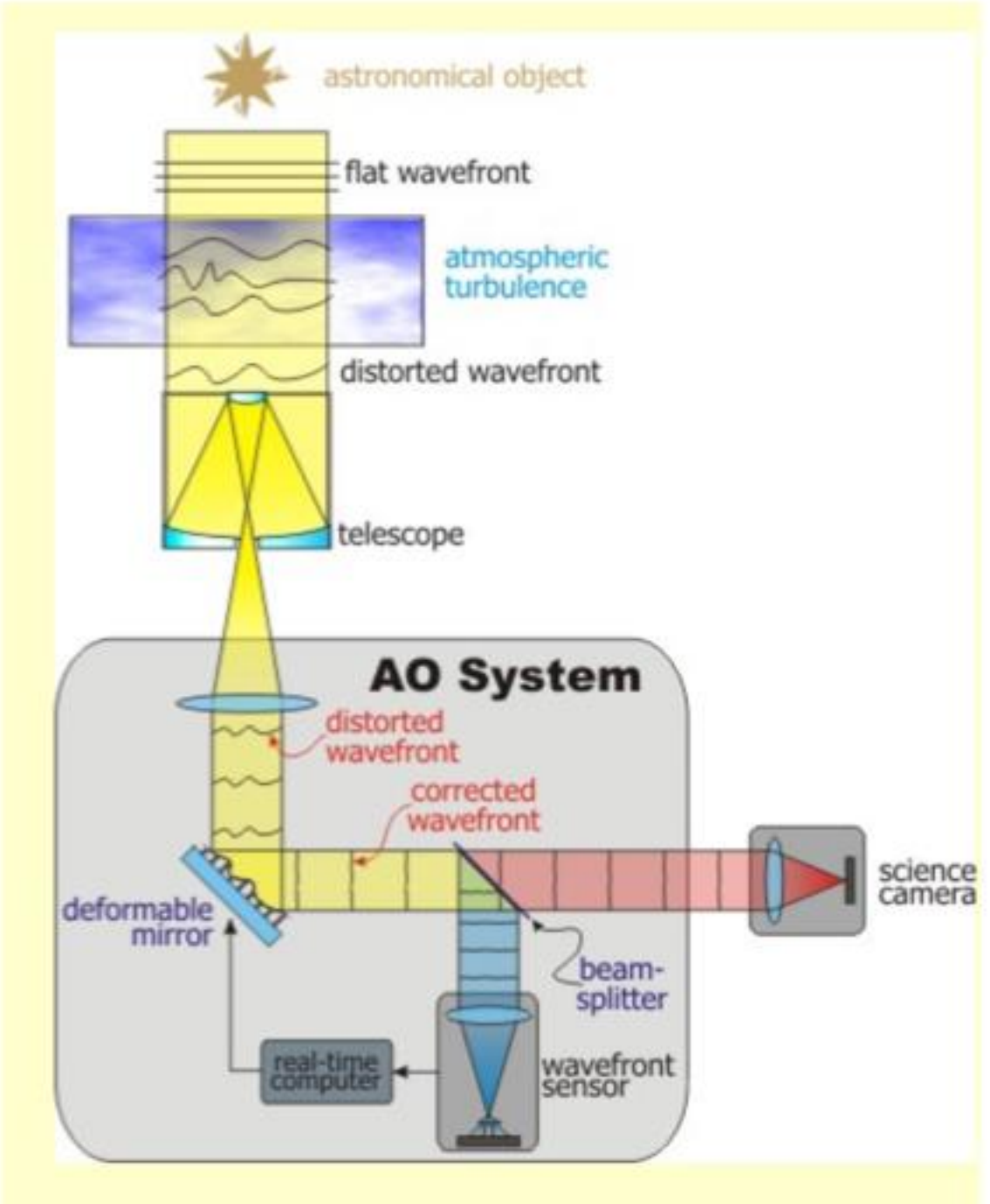
Dealing with Turbulence: Adaptive Optics



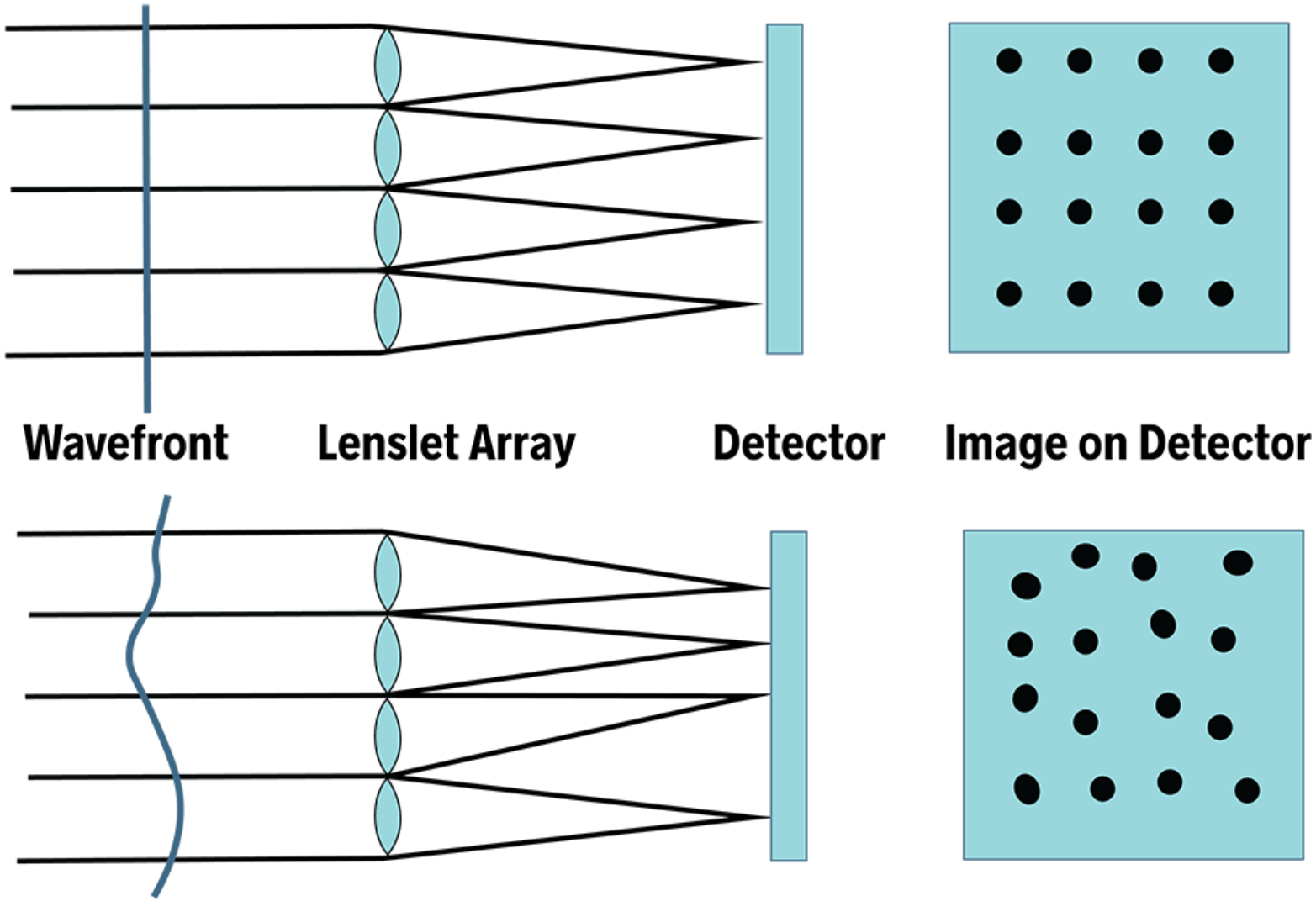


Adaptive Optics Application

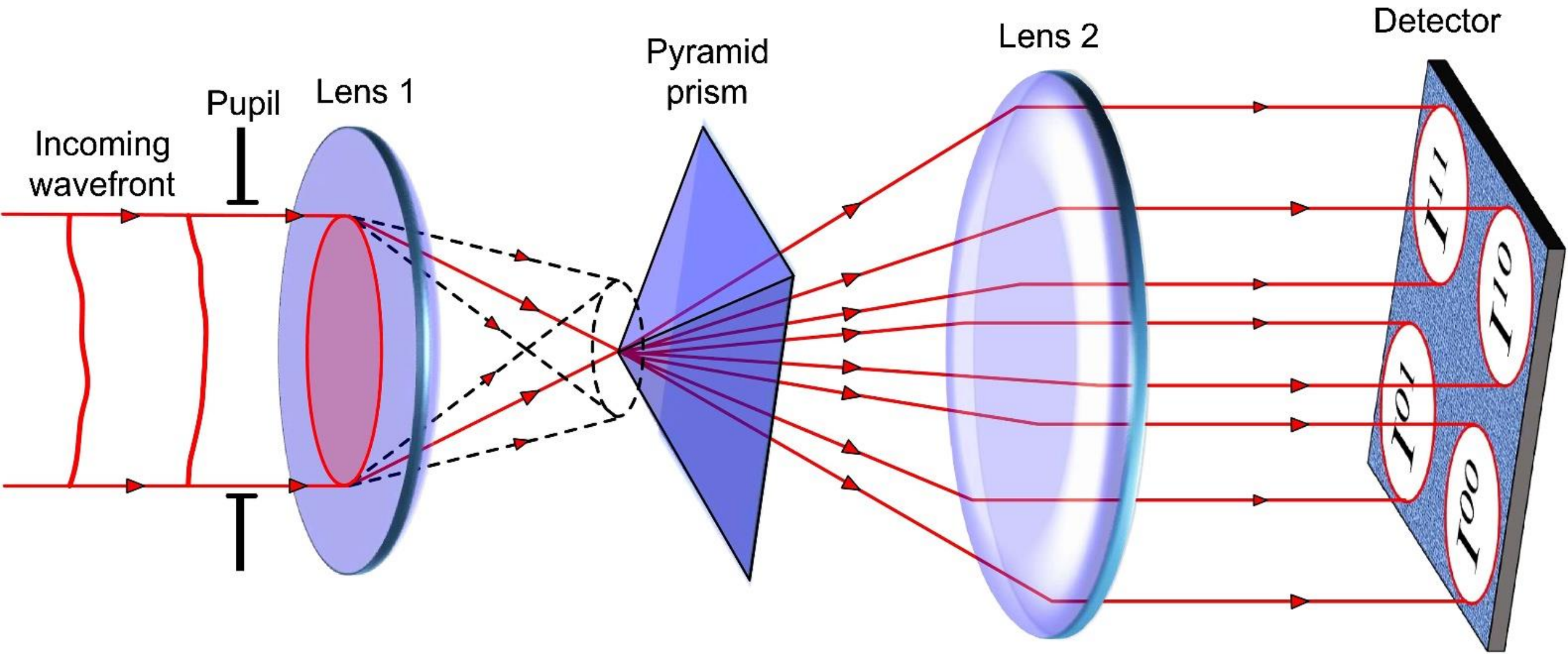
- Real time (<100 ms update rate) correction of wavefront aberrations induced by atmospheric turbulence.
- Requires a relatively bright (V<15) reference star although artificial beacons are in use.
- In practice you need observers to run the AO system and the instrument. Some AO systems are labor intensive.



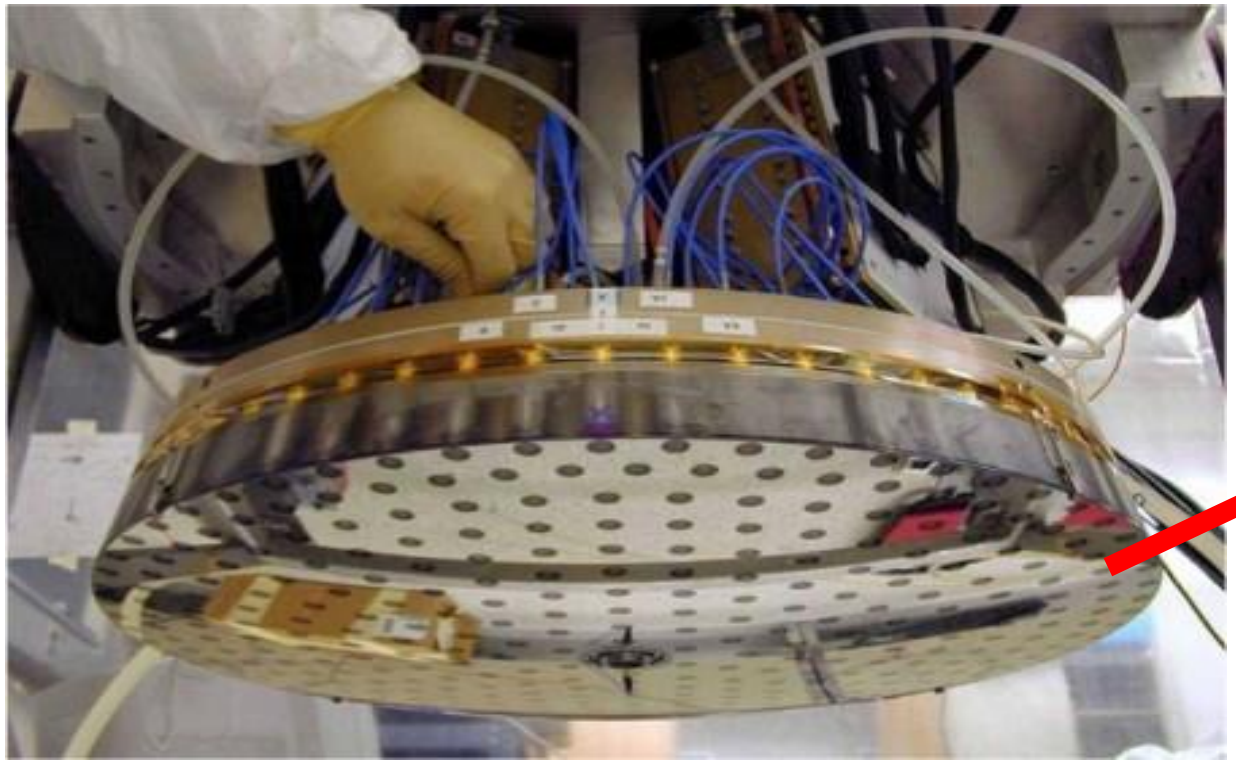
Shack-Hartmann Wavefront Sensor

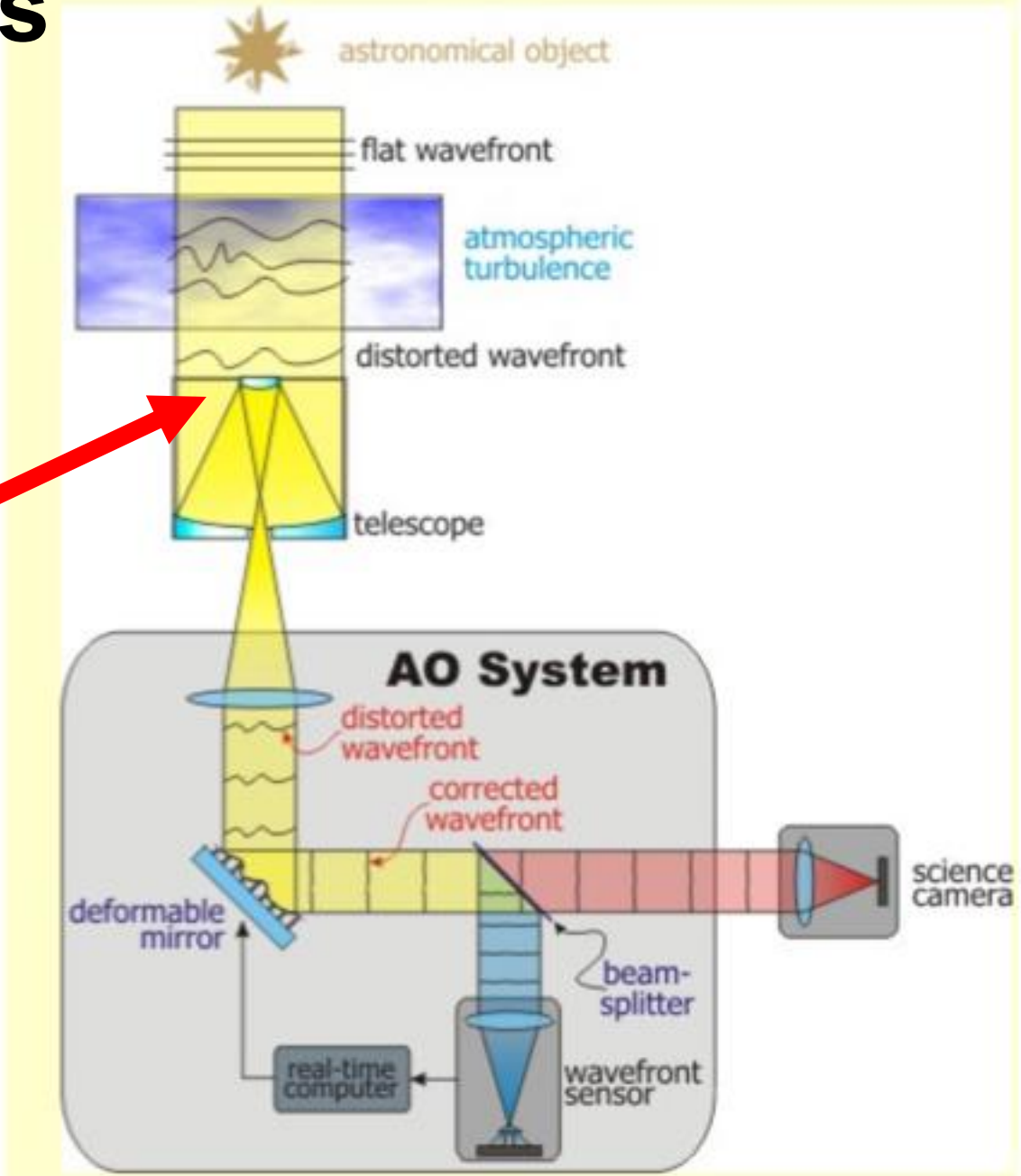


Pyramid Wavefront Sensor



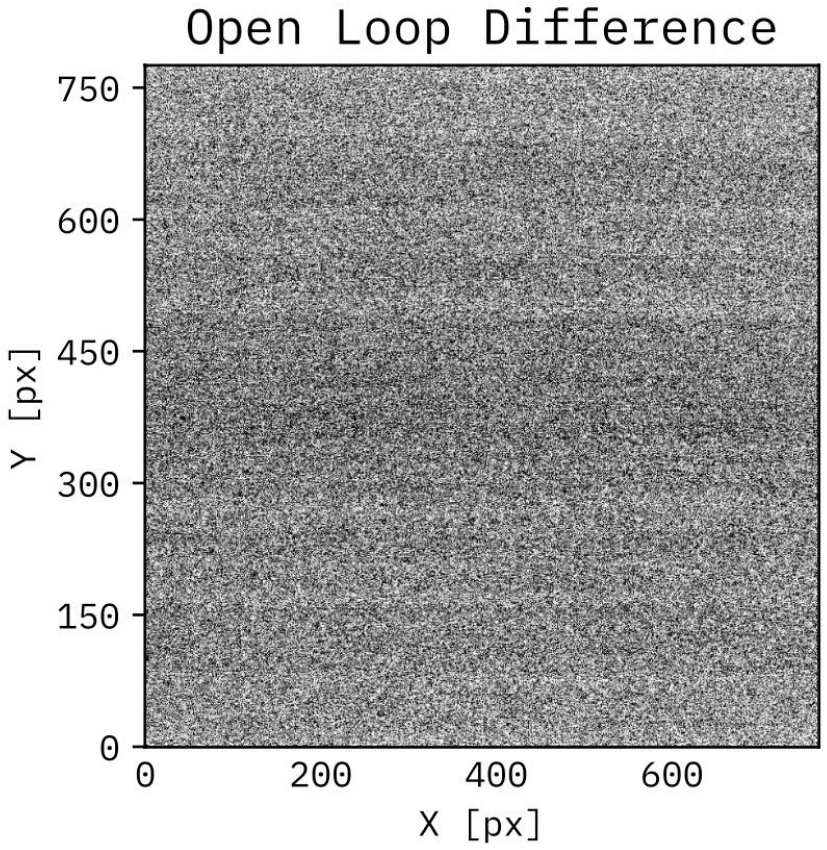
Adaptive Secondary Mirrors

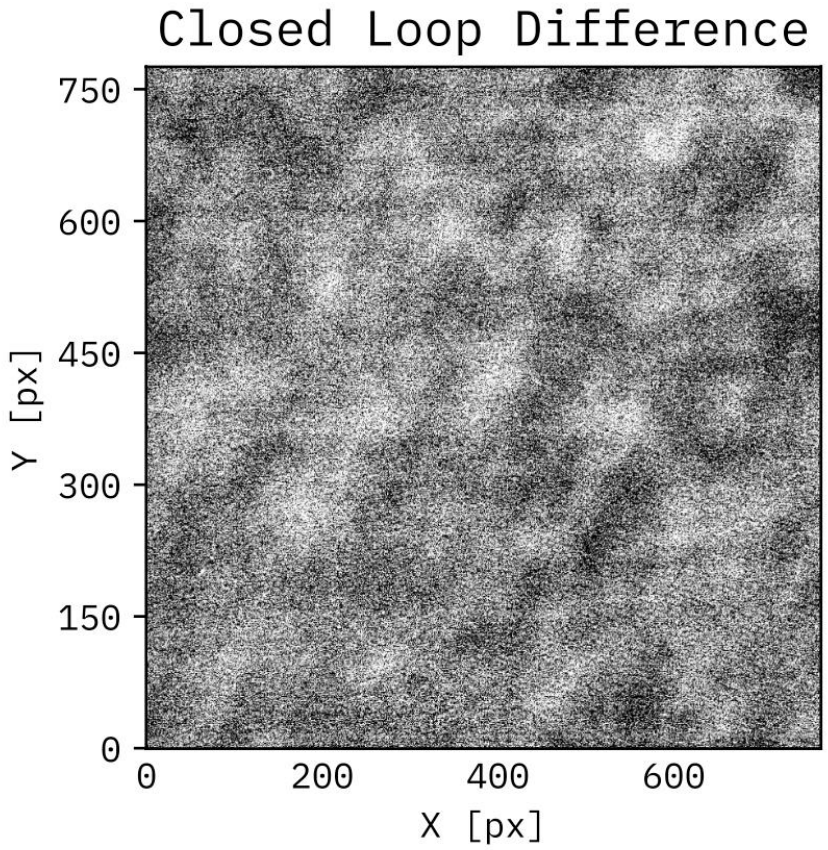




- The smaller mirrors of a telescope can be equipped with actuators or pistons to correct for atmospheric turbulence.

Why are Adaptive Secondaries Necessary?





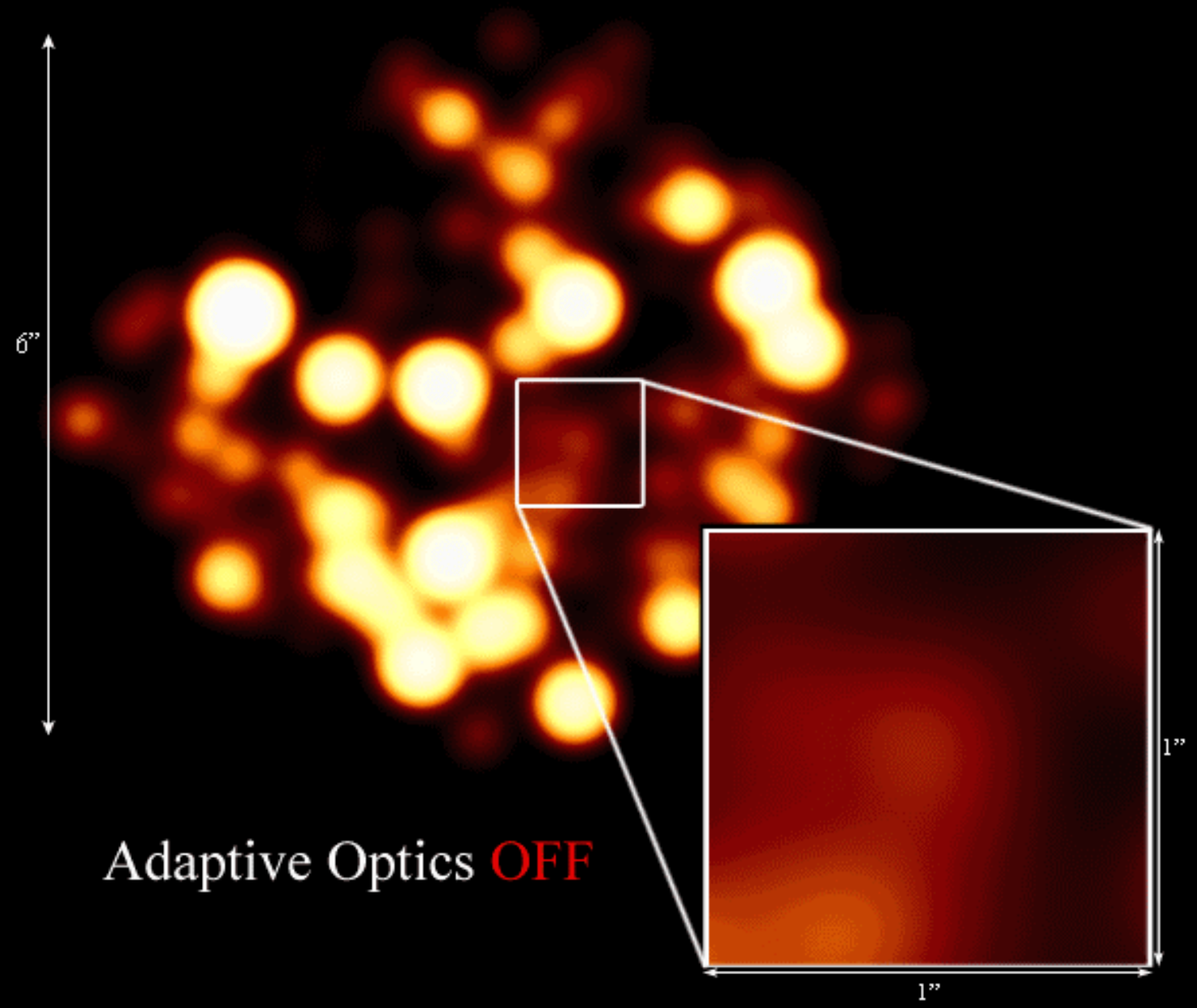
Keck/NIRC2
Nguyen+ 2025

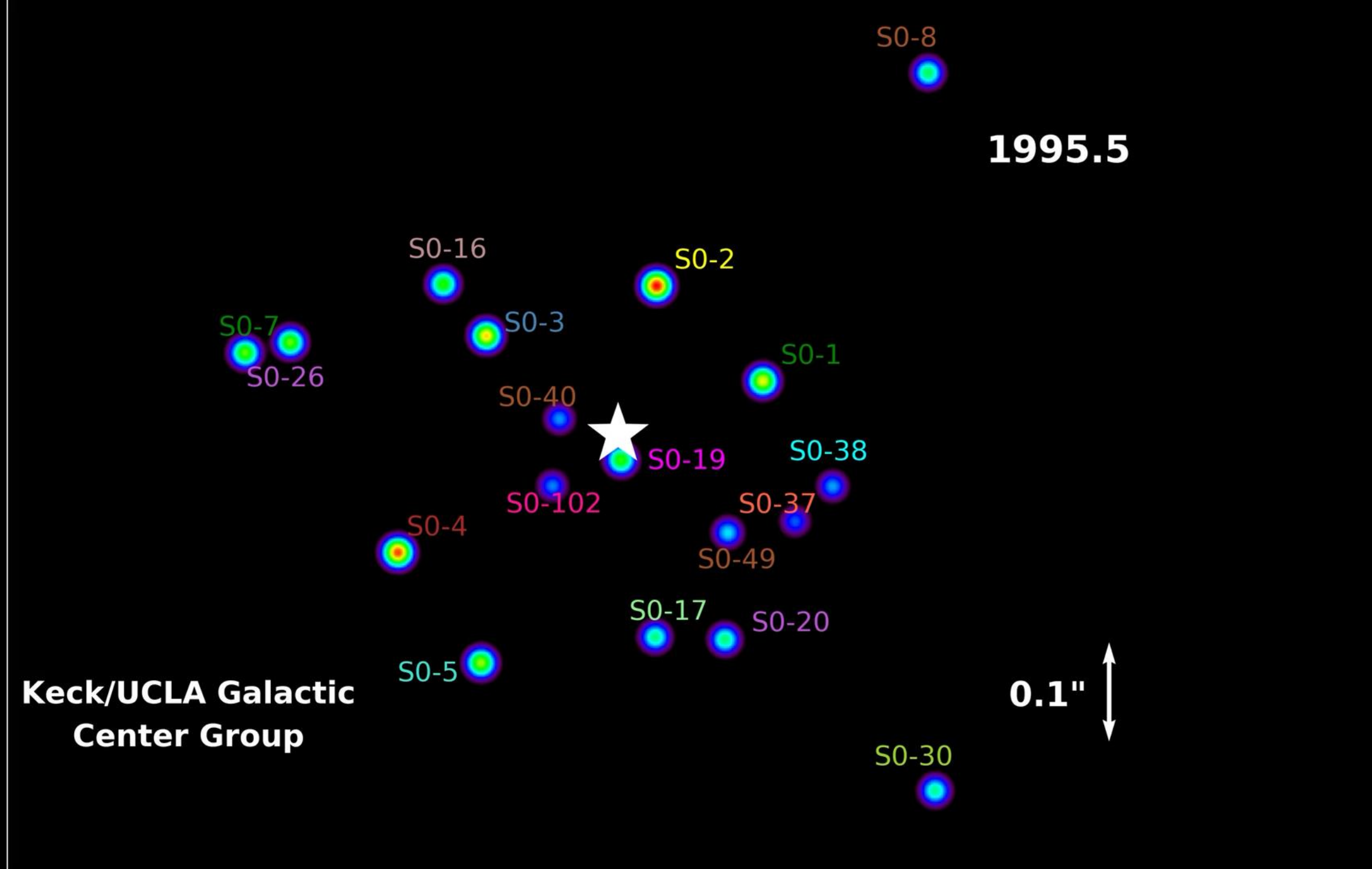
Adaptive secondaries limit thermal background noise compared to standard AO systems.


Adaptive optics

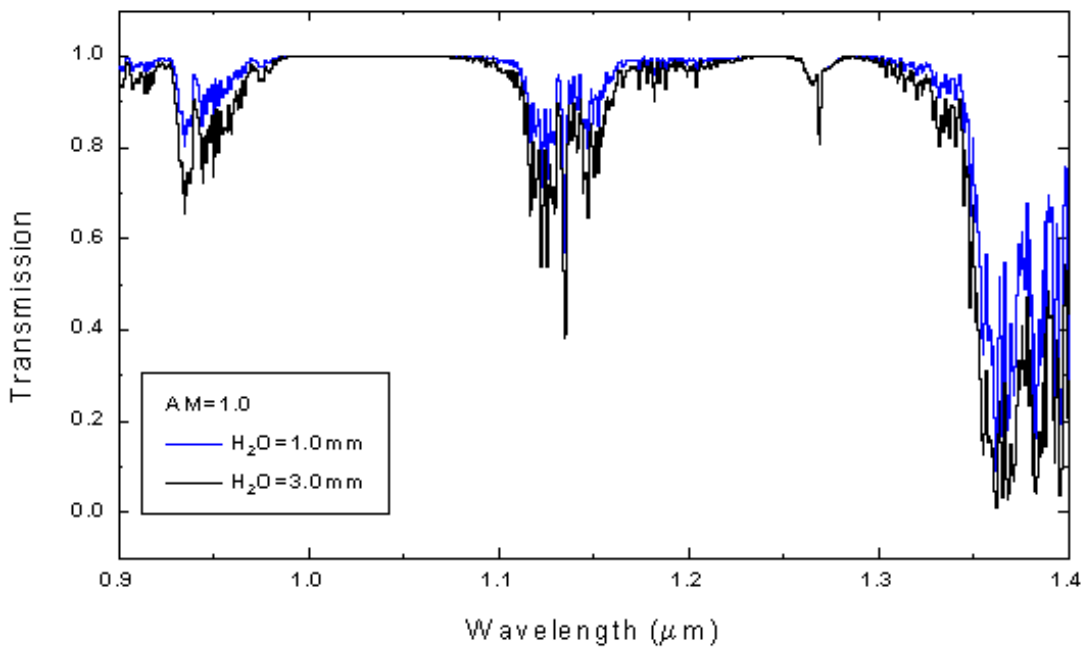

No Adaptive optics

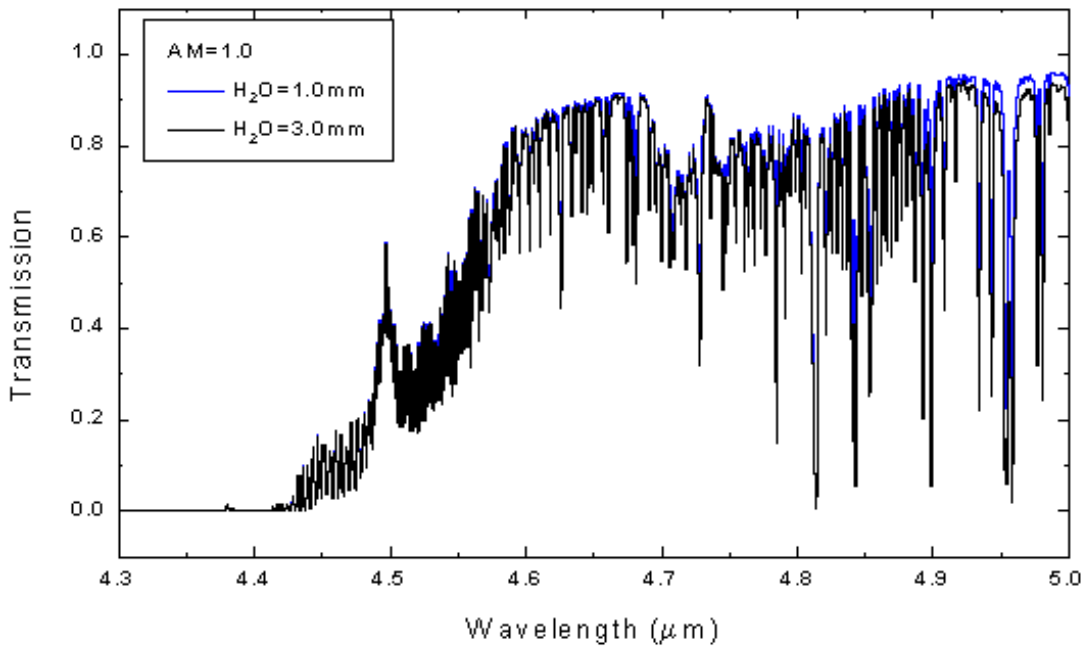
The Galactic Center at 2.2 microns




S0-8
1995.5
S0-16
S0-2
S0-7
S0-3
S0-1
S0-26
S0-40
S0-19
S0-38
S0-102
S0-37
S0-4
S0-49
S0-17
S0-20
S0-5
0.1"
Keck/UCLA Galactic
Center Group
S0-30

The Impact of Earth's Atmosphere: Water Vapor

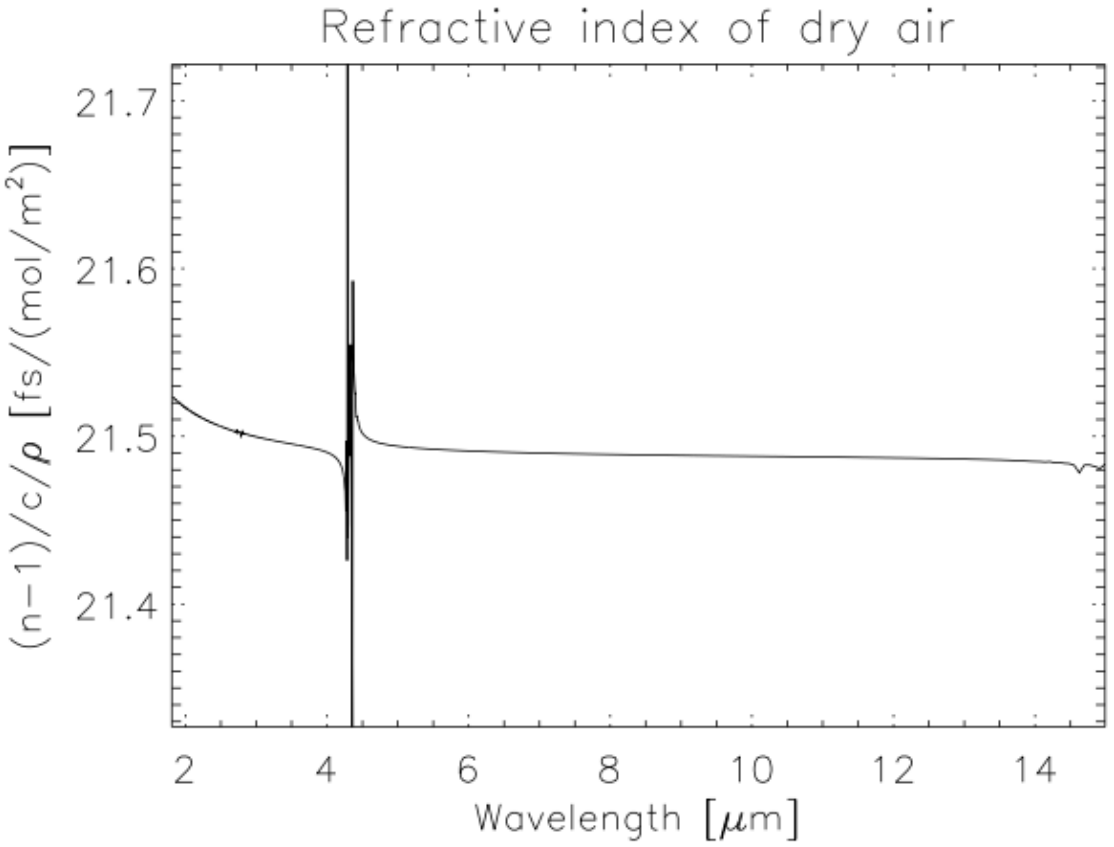


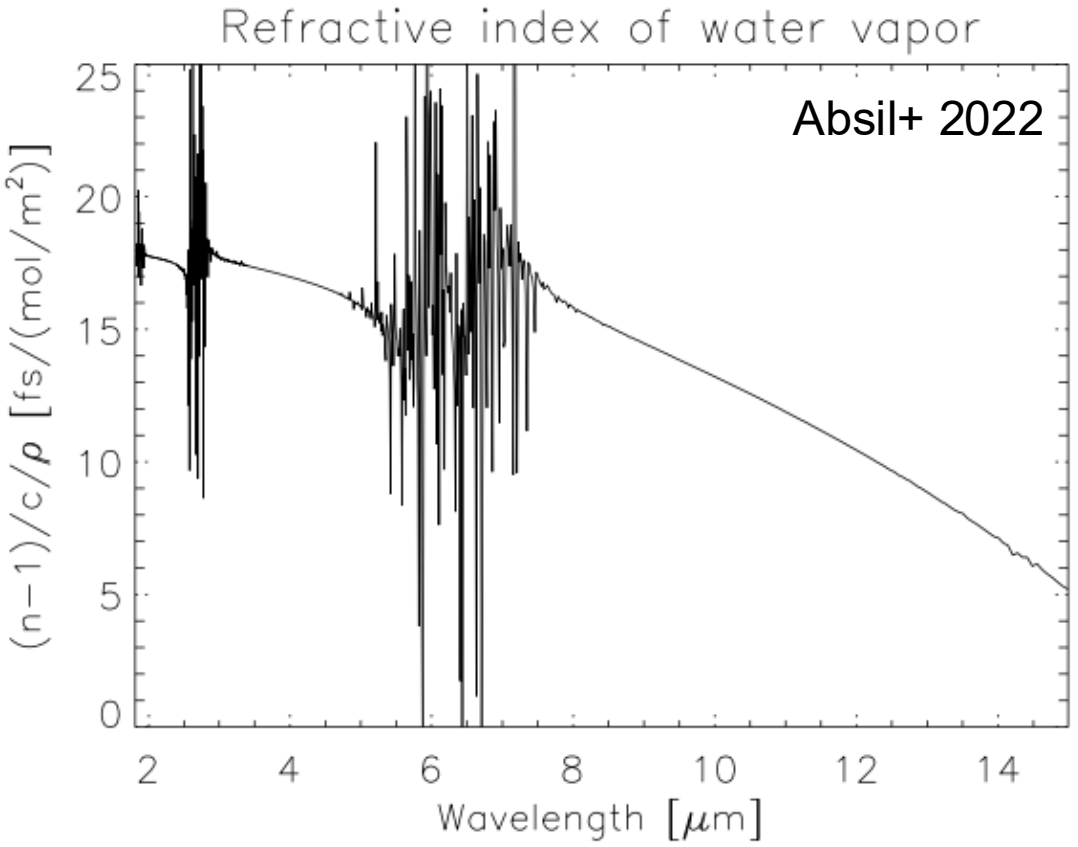


- Precipitable Water Vapor (**PWV**) is a measure of how much moisture is in the air if it were to condense out into a column.

- Water vapor column behavior is seasonal and is also improved by going to higher elevations.

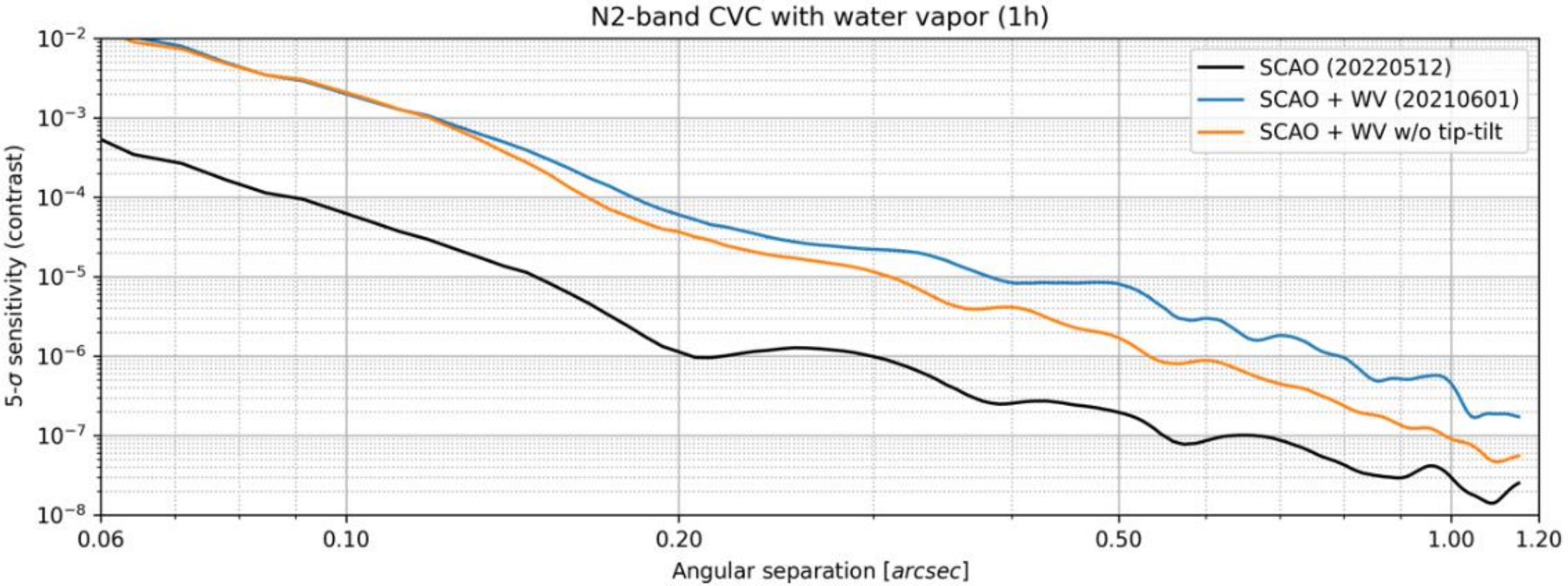
The Impact of Earth's Atmosphere: Water Vapor





- The refractive index of dry air is relatively flat vs wavelength.
- Water vapor is very chromatic, especially at longer-wavelengths.
- If the behavior of water is very different than the air it negatively impacts observations.

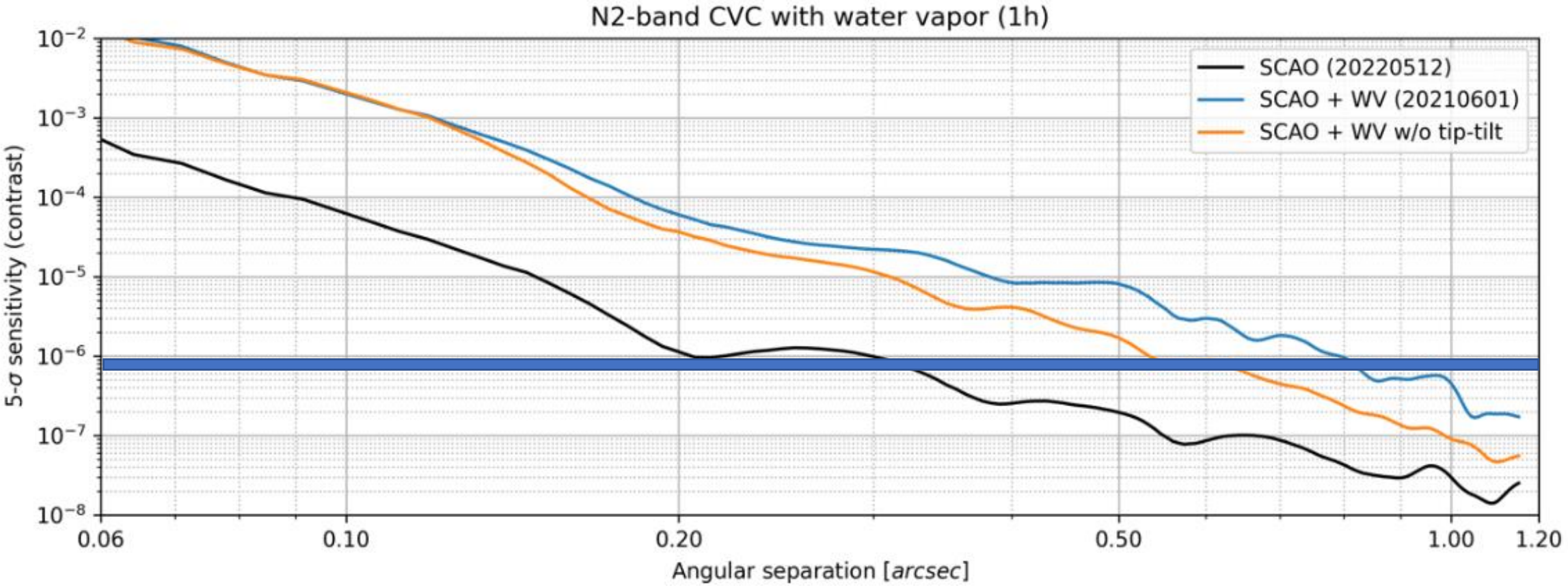
Water Vapor and Planet Detection



Contrast or “detectability” at N-band (8-13 microns). Lower contrast values means you can detect dimmer planets.

Absil+ 2024

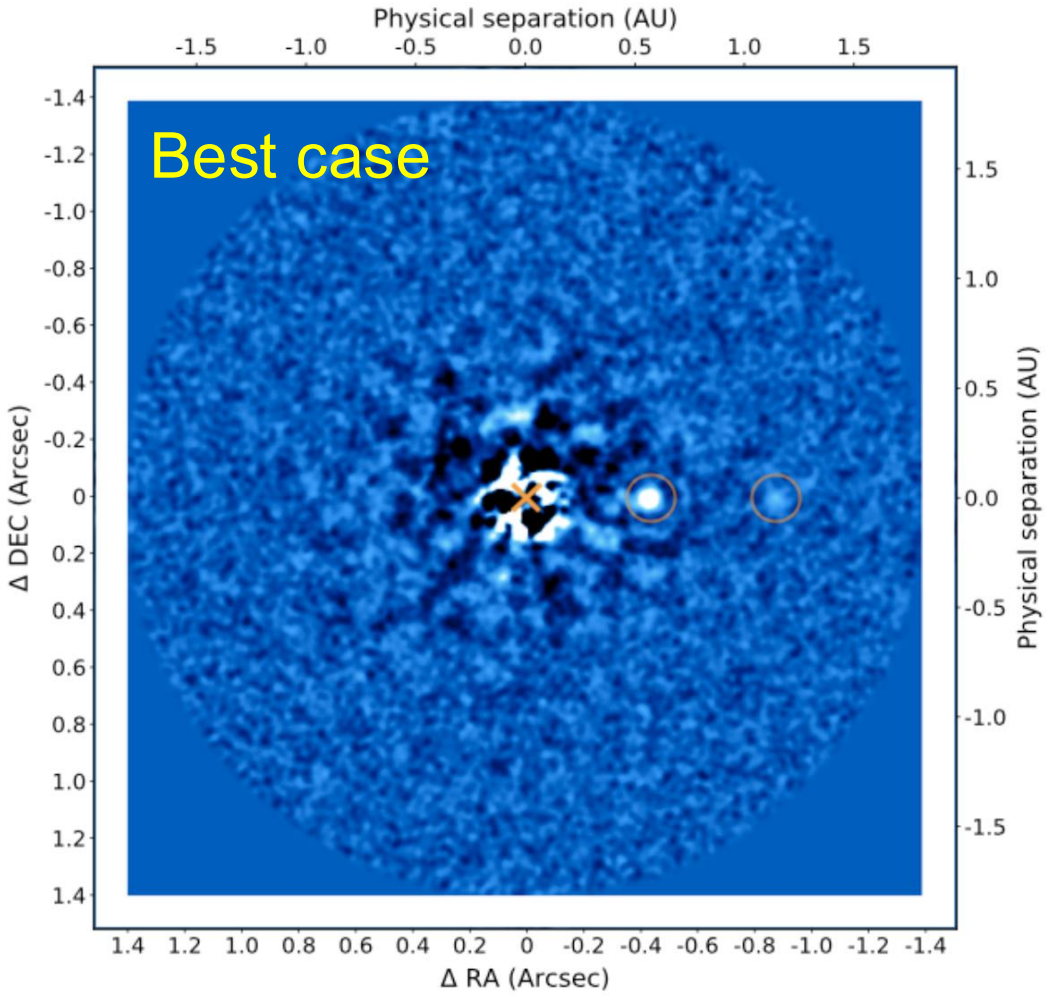
Water Vapor and Planet Detection

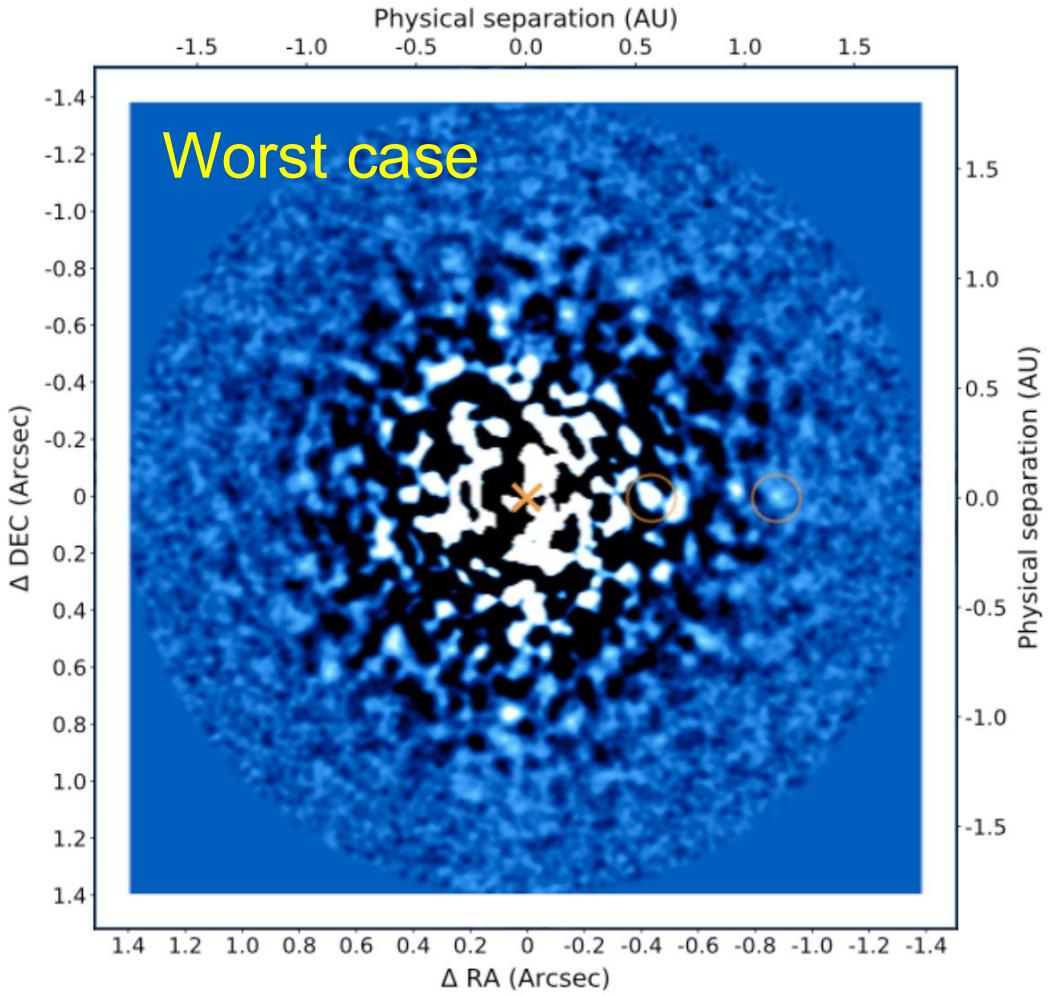


Contrast or “detectability” at N-band (8-13 microns). Lower contrast values means you can detect dimmer planets.

Absil+ 2024

Water Vapor and Planet Detection





Simulation of Earth analog around Alpha Cen with ELT/METIS (30-meter) at N-band (8-13 microns)

Absil+ 2024